From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT –IV
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A, B, C	D
2.	C	A, C	C
3.	D	C	A
4.	B	A, B, D	B
5.	B	B, C, D	A, C
6.	C	A, C	A, C, D
7.	C	A, B	A, B, D
8.	C	A, B, C	B, D
9.	C	B	B
10.	A	D	C
11.	B	B	B
12.	B	B	D
13.	C	B	A
14.	B	D	B
15.	A	A	D
16.	C	B	C
17.	C	C	B
18.	D	A	C
19.	C	D	D
20.	B	C	C

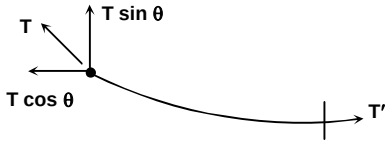
Physics

PART – I

2. $T \sin\theta = mg/2$

$T' = T\cos\theta$

$= \frac{mg}{2}\cot\theta = \frac{mg}{2\tan\theta}$



3. $\frac{1}{2}mv^2 = pt$ (P = const)

$\Rightarrow v = \sqrt{\frac{2Pt}{m}}$

$a = \frac{dv}{dt} = \sqrt{\frac{2P}{m}}\frac{1}{2\sqrt{t}}$

$F = ma = \sqrt{\frac{mP}{2t}} \propto \frac{1}{v}$

5. $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5$

$\left|\vec{F}_2\right| = \left|\vec{F}_5\right|$ and $\left|\vec{F}_2\right| = \left|\vec{F}_4\right|$

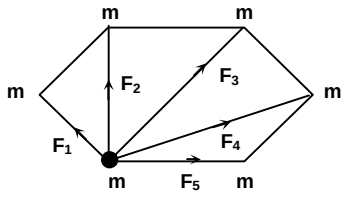
$\vec{F}_1 = \vec{F}_3 + 2F_2\cos 30° + 2F_1\cos 60°$

$F_3 = \frac{Gm^2}{4a^2}$; $F_2 = \frac{Gm^2}{3a^2}$; $F_1 = \frac{Gm^2}{a^2}$

$F = \frac{Gm^2}{a^2}\left(\frac{5}{4} + \frac{1}{\sqrt{3}}\right) = m\omega^2 a$

$\omega = \sqrt{\frac{Gm}{a^3}\left(\frac{5}{4} + \frac{1}{\sqrt{3}}\right)}$

$T = 2\pi\sqrt{\frac{4\sqrt{3}a^3}{Gm\left(5\sqrt{3}+4\right)}}$



6. $\rho + \rho g(L - x) - \rho gL = \frac{d}{dt}(\rho xu)$

7. $v_{p_x} = 5$, $v_{p_y} = 5$

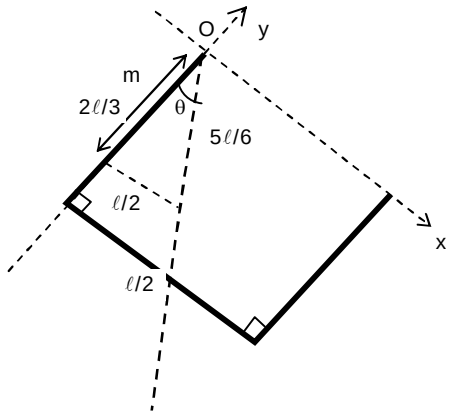
$5\sqrt{2}$ = 7.1 m/s approx.

8. $v_P = \frac{1600\times10^{-6}}{4\times10^{-4}}$ = 4 m/s

$v_Q = \frac{1600\times10^{-6}}{2\times10^{-4}}$ = 8 m/s

$P_A - P_B = \frac{1}{2}\rho_A\left(v_Q^2 - v_P^2\right) = \rho gh$

11-12. Let the point of suspension is origin(O), then C.M. lies on vertical line when it is in equilibrium position.



$x_{cm} = \dfrac{m(0)+m(L/2)+m(L)}{3m} = L/2$ and

$y_{cm} = \dfrac{m(-L/2)+m(-L)+m(-L/2)}{3m} = \dfrac{-2L}{3}$

$\therefore \quad \tan\theta = \dfrac{L/2}{2L/3} = ¾ \Rightarrow \theta = \tan^{-1}(3/4)$ or 37^0

$d^2 = \left(\dfrac{2L}{3}\right)^2 + \left(\dfrac{L}{2}\right)^2$

$d = \dfrac{5L}{6}$

Moment of inertia $I = Mk^2 = 4m\left(\dfrac{\ell^2}{12}+\dfrac{\ell^2}{4}+\dfrac{\ell^2}{2}\right) - \dfrac{m\ell^2}{3} = 3m\ell^2$

(Assume full square and calculate moment of inertia about origin and subtract the moment of inertia of fourth rod.)

Time period of a physical pendulum $= 2\pi\sqrt{\dfrac{I}{mgd}} = 2\pi\sqrt{\dfrac{6\ell}{5g}}$ sec

13-14. $F = qvB \sin\theta = qvB \ (\theta = 90°)$

So, $B = \dfrac{F}{qv} = \dfrac{3.2\times10^{-20}}{1.6\times10^{-19}\times4\times10^{3}} = 5\times10^{-7}$ T

Now, $\dfrac{\mu_0}{2\pi}\left[\dfrac{I_p}{5}+\dfrac{I_q}{2}\right] = 5\times10^{-7}$

$\Rightarrow$ I = 4 amp.

If the distance of point R from third current carrying current is X, then

$B_R = 0$

$\dfrac{\mu_0}{4\pi}\dfrac{2\times2.5}{|x|} + 5\times10^{-7} = 0$

so $x = \pm 1$ m

16. $P(2\pi r)dr = 2\pi r\ell\eta \dfrac{dv}{dr} - 2\pi(r+dr)\,\ell\eta \dfrac{dv}{dr}$

$P(2\pi r)dr = -2\pi\,\ell\eta\left(\dfrac{dv}{dr}\right)dr$

$P(2\pi r) = -2\pi\,\ell\eta\left(\dfrac{-2v_0 r}{R^2}\right)$

$P = \dfrac{2\eta\ell v_0}{R^2}$

Chemistry

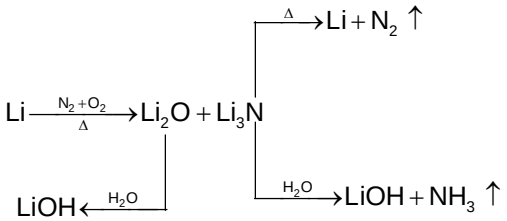
PART – II

SECTION – A

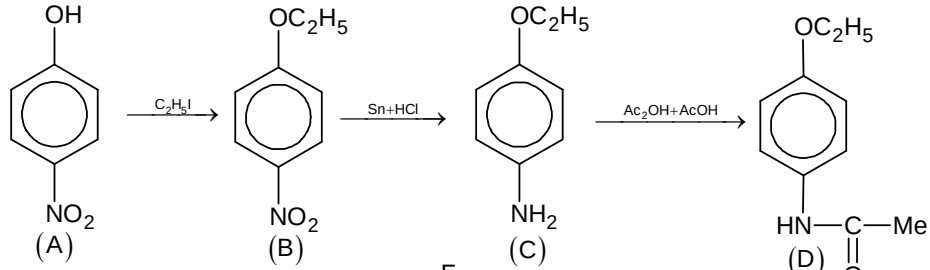
8. (D) is incorrect, since Gabriel method is used to prepare 1° aliphatic amines.

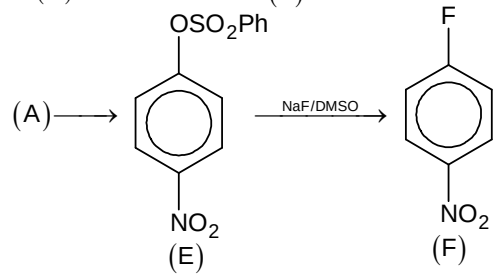
10. Bottles 1, 2, 3, 4 have $Pb(NO_3)_2$, HCl, Na_2CO_3 and $CuSO_4$.
When mixing:
Bottle 1 + Bottle 2 $\rightarrow PbCl_2$
Bottle 1 + Bottle 3 $\rightarrow PbCO_3$
Bottle 1 + Bottle 4 $\rightarrow PbSO_4$
Bottle 2 + Bottle 3 $\rightarrow CO_2$
Bottle 2 + Bottle 4 $\rightarrow$ No visible reaction
Bottle 3 + Bottle 4 $\rightarrow CuCO_3$

12.



Solution for the Q. No. 15 & 16





20. (P – 2) oxalic acid dehydrate, loses H_2O when treated at $105^{o}C$.
(Q – 3) when heated at $200^{o}C$ or when refluxed with H_2SO_4 at $90^{o}C$, it decomposes into CO_2, CO, HCOOH and H_2O.
(R – 4) malonic acid on heating with P_2O_5 loses H_2O and forms carbon suboxide.
(S – 1) oxalic acid is oxidised by $KMnO_4$ to CO_2.

Mathematics

PART – III

HINTS AND SOLUTION

1. Put $x^2 = t$

$$I = \frac{1}{2}\int e^t\left[te^{t^2} + \left(e^{t^2} + 2t^2e^{t^2}\right)\right] dt$$

2. We have $\frac{a}{2} = \frac{b}{3} = k$ and $\sec^2 A = \frac{8}{5}$

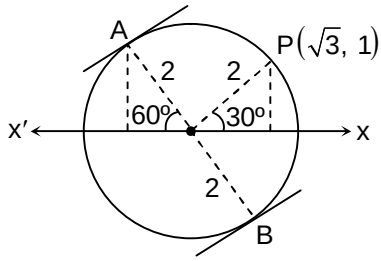
$\Rightarrow \cos^2 A = \frac{5}{8}$

$\Rightarrow c^2 = 10k^2,\ \frac{5}{2}k^2$

4. $$\tan\left(\theta_1 + \theta_2 + \theta_3\right) = \frac{\sum\tan\theta_1 - \prod\tan\theta_1}{1 - \sum\tan\theta_1\tan\theta_2} = \frac{a + 1 - b}{1 - (b - a)} = 1$$

7. $$f(x) = \begin{cases} -x & \text{if } x \in [1, 2) \\ x & \text{if } x \in (2, \infty) \end{cases}$$

8. Clearly, A = $(-2\cos 60^\circ, 2\sin 60^\circ)$
B = $(2\cos 60^\circ, -2\sin 60^\circ)$
Tangent at A is $x(-2\cos 60^\circ) + y(2\sin 60^\circ) = 4$
and tangent at B is $x(2\cos 60^\circ) + y(-2\sin 60^\circ) = 4$



9. Since, parabola is above x–axis
$D = a^2 - 4 \le 0$
Thus, $a_{max} = 2$

10. Equation of tangent at (0, 1) to the parabola is $y - 1 = a(x - 0)$

11.-12. $az + b\bar{z} + c = 0$ ..... (1)

$\Rightarrow \bar{a}\bar{z} + \bar{b}z + \bar{c} = 0$ ..... (2)

Eliminating $\bar{z}$ from (1) and (2)

$\left(|b|^2 - |a|^2\right)z = c\bar{a} - b\bar{c}$

Adding (1) and (2) $(\bar{a} + b)\bar{z} + (a + \bar{b})z + (c + \bar{c}) = 0$

This is of form $A\bar{z} + \bar{A}z + B = 0$

13.-14 Here, $e^A = \frac{1}{2}\begin{bmatrix} e^{2x} + 1 & e^{2x} - 1 \\ e^{2x} - 1 & e^{2x} + 1 \end{bmatrix}$

So, $f(x) = e^{2x} + 1$; $g(x) = e^{2x} - 1$

15.-16. $f(x) - \lambda \sin x \int_0^{\pi/2} \cos t\, f(t)\,dt = \sin x$

$\Rightarrow$ f(x) – P sin x = sin x

or f(x) = (P + 1) sin x

Where $P = \lambda \int_0^{\pi/2} \cos t\, f(t)\,dt$

$= \lambda \int_0^{\pi/2} \cos t (P+1) \sin t\, dt$

$= \frac{\lambda(P+1)}{2} \int_0^{\pi/2} \sin 2t\, dt$

$P = \frac{\lambda(P+1)}{2} \Rightarrow P = \frac{\lambda}{2-\lambda}$

$f(x) = \left(\frac{\lambda}{2-\lambda} + 1\right) \sin x = 2$

$\Rightarrow \sin x = 2 - \lambda$

$\Rightarrow \lambda \in [1, 3]$

$\int_0^{\pi/2} \frac{2}{2-\lambda} \sin x\, dx = 3$

$\Rightarrow \lambda = \frac{4}{3}$

20. (P) $2(F(x) - f(x)) = f^2(x)$

$\Rightarrow f(x) - f'(x) = f(x)\, f'(x)$

$\Rightarrow f'(x) = \frac{f(x)}{1+f(x)} = 1 - \frac{1}{1+f(x)}$

$\Rightarrow$ f(x) is an increasing function

(Q) $21 \le f(5) - f(-2) \le 28$

(R) $y = \frac{x+2}{2x^2+3x+6}$

$\Rightarrow y \in \left[-\frac{1}{13}, \frac{1}{3}\right]$

(S) $y = e^{-x} \cos x$

$\Rightarrow y_1 = -e^{-x} \sin x - e^{-x} \cos x = -e^{-x} \sin x - y$

$y_2 = -e^{-x} \cos x + e^{-x} \sin x - y_1$

$\Rightarrow y_4 + 4y = 0$

$\Rightarrow k = 2$